Version 7.4.2

PVsyst - Simulation report

Grid-Connected System

Project: Kebithigollewa 10MWac Fixed Tilt - Huawei 300kW-Astronergy 580W

Variant: Huawei 300KTL Astronergy 580W N

No 3D scene defined, no shadings

System power: 13.51 MWp

Kebithigollewa - Sri Lanka

Author

Windforce PLC (Sri Lanka)

PVsyst V7.4.2VC0,
Simulation date:
11/11/23 09:53
with v7.4.2

Windforce PLC (Sri Lanka)

Project summary

Geographical Site
Kebithigollewa
Sri Lanka

Situation

Latitude	8.56 °N
Longitude	80.66 °E
Altitude	123 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Meteo data
Kebithigollewa
Meteonorm 8.1, Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane	
Tilt/Azimuth	8 / 0 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	23296 units
Pnom total	13.51 MWp

Inverters

Nb. of units	32 units
Pnom total	9600 kWac
Grid power limit	10000 kWac
Grid lim. Pnom ratio	1.351

Results summary

Produced Energy	21881645 kWh/year	Specific production	1619 kWh/kWp/year	Perf. Ratio PR	86.95 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		**Sheds configuration**	**Models used**	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	8 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon	**Near Shadings**	**User's needs**
Free Horizon	No Shadings	Unlimited load (grid)

Grid power limitation

Active power	10000 kWac
Pnom ratio	1.351

PV Array Characteristics

PV module

Manufacturer	Astronergy
Model	CHSM72N(DG)/F-HC-580
(Custom parameters definition)	
Unit Nom. Power	580 Wp
Number of PV modules	23296 units
Nominal (STC)	13.51 MWp
Modules	832 Strings x 28 In series
At operating cond. (50°C)	
Pmpp	12.55 MWp
U mpp	1045 V
I mpp	12008 A

Inverter

Manufacturer	Huawei Technologies
Model	SUN2000-330KTL-H1
(Custom parameters definition)	
Unit Nom. Power	300 kWac
Number of inverters	32 units
Total power	9600 kWac
Operating voltage	500-1500 V
Max. power (=>30°C)	330 kWac
Pnom ratio (DC:AC)	1.41
Power sharing within this inverter	

Total PV power

Nominal (STC)	13512 kWp
Total	23296 modules
Module area	60179 m²
Cell area	55385 m²

Total inverter power

Total power	9600 kWac
Max. power	10560 kWac
Number of inverters	32 units
Pnom ratio	1.41

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	1.4 mΩ
Loss Fraction	1.5 % at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

Loss Fraction	2.0 % at MPP

Strings Mismatch loss

Loss Fraction	0.2 %

IAM loss factor

Incidence effect (IAM): User defined profile

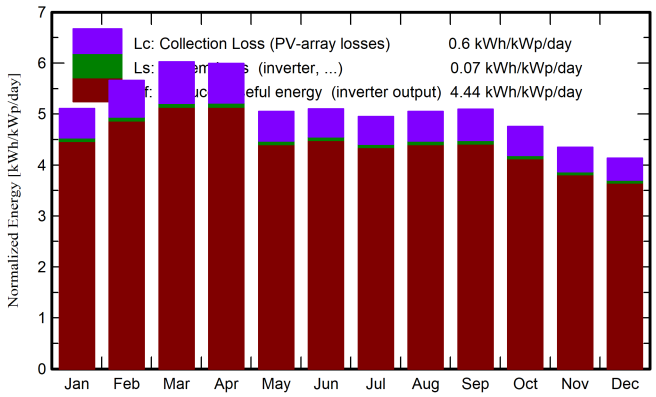
0°	40°	50°	60°	70°	75°	80°	85°	90°
1.000	1.000	1.000	1.000	1.000	0.984	0.949	0.830	0.000

Main results

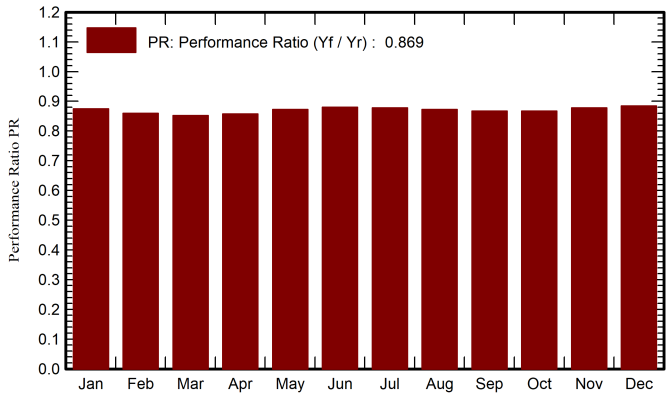
System Production

Produced Energy (P50)	21881645 kWh/year	Specific production (P50)	1619 kWh/kWp/year	Perf. Ratio PR	86.95 %
Produced Energy (P90)	20237360 kWh/year	Specific production (P90)	1498 kWh/kWp/year		
Produced Energy (P95)	19774722 kWh/year	Specific production (P95)	1464 kWh/kWp/year		

Normalized productions (per installed kWp)



Performance Ratio PR



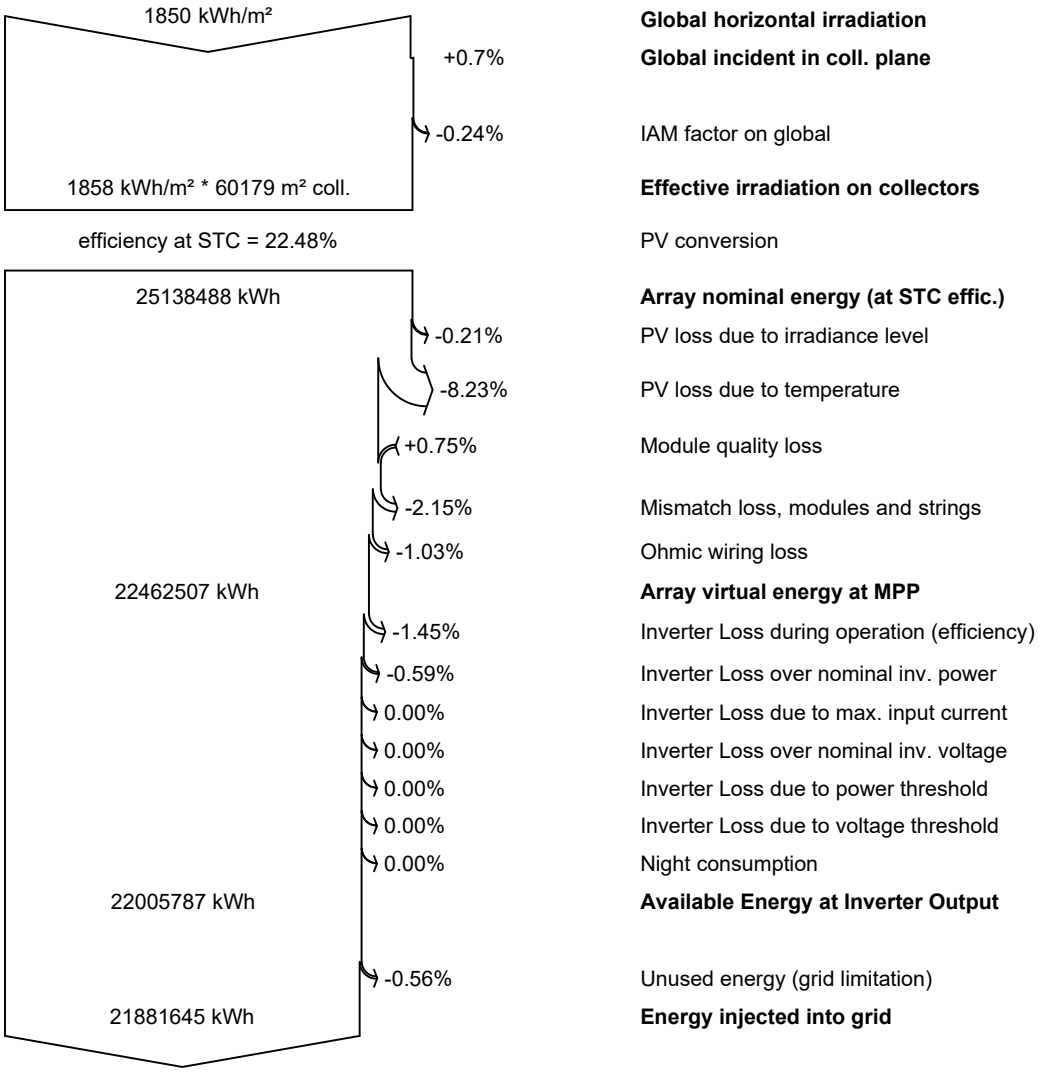
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	148.6	70.90	25.92	158.3	158.0	1900354	1872138	0.875
February	151.5	65.97	26.77	158.5	158.2	1870074	1841873	0.860
March	183.7	77.83	28.12	186.8	186.5	2185580	2152112	0.852
April	182.1	78.82	28.70	179.9	179.5	2115476	2083129	0.857
May	162.1	83.47	29.70	156.7	156.2	1874500	1846367	0.872
June	160.1	82.46	29.31	153.1	152.6	1847185	1819886	0.880
July	159.8	83.49	29.39	153.6	153.1	1849105	1821661	0.878
August	159.8	82.77	28.90	156.6	156.2	1874341	1846115	0.873
September	152.6	81.31	28.38	153.0	152.6	1818772	1791433	0.867
October	143.8	80.20	27.76	147.5	147.2	1754739	1728582	0.867
November	124.9	73.80	26.37	130.4	130.1	1570506	1547564	0.878
December	120.9	66.48	26.11	128.2	127.9	1553617	1530786	0.884
Year	1849.9	927.49	27.96	1862.6	1858.2	22214252	21881645	0.869

Legends

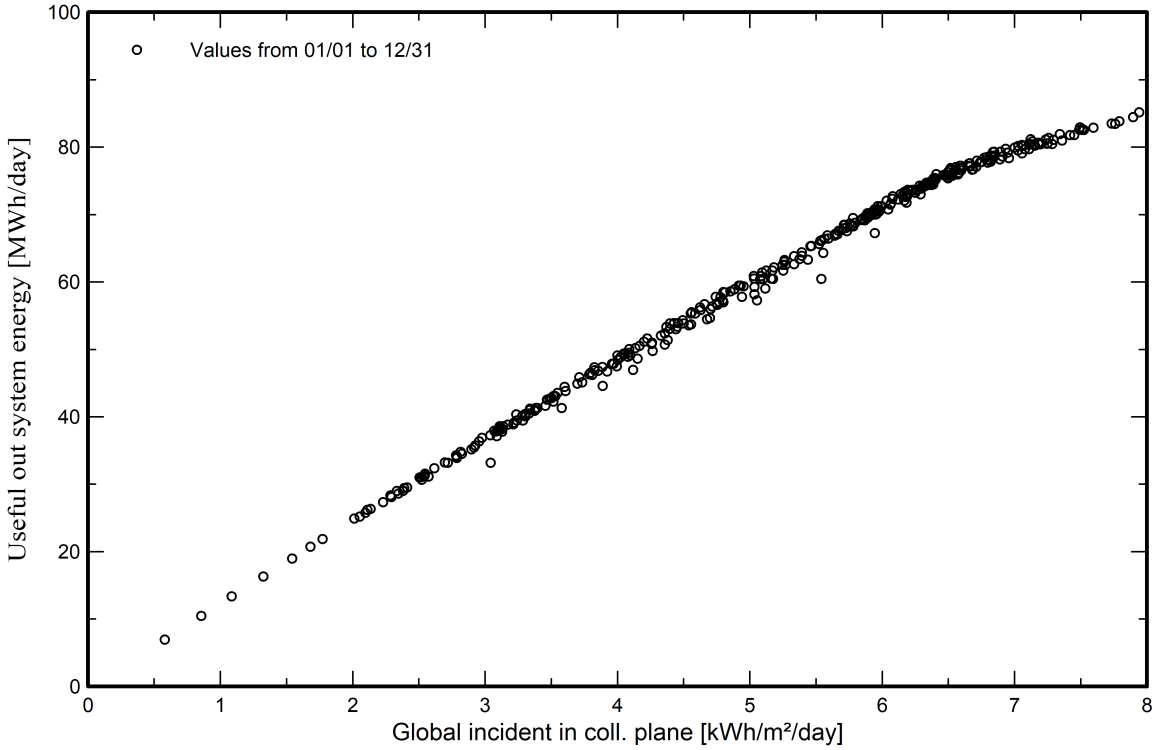
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

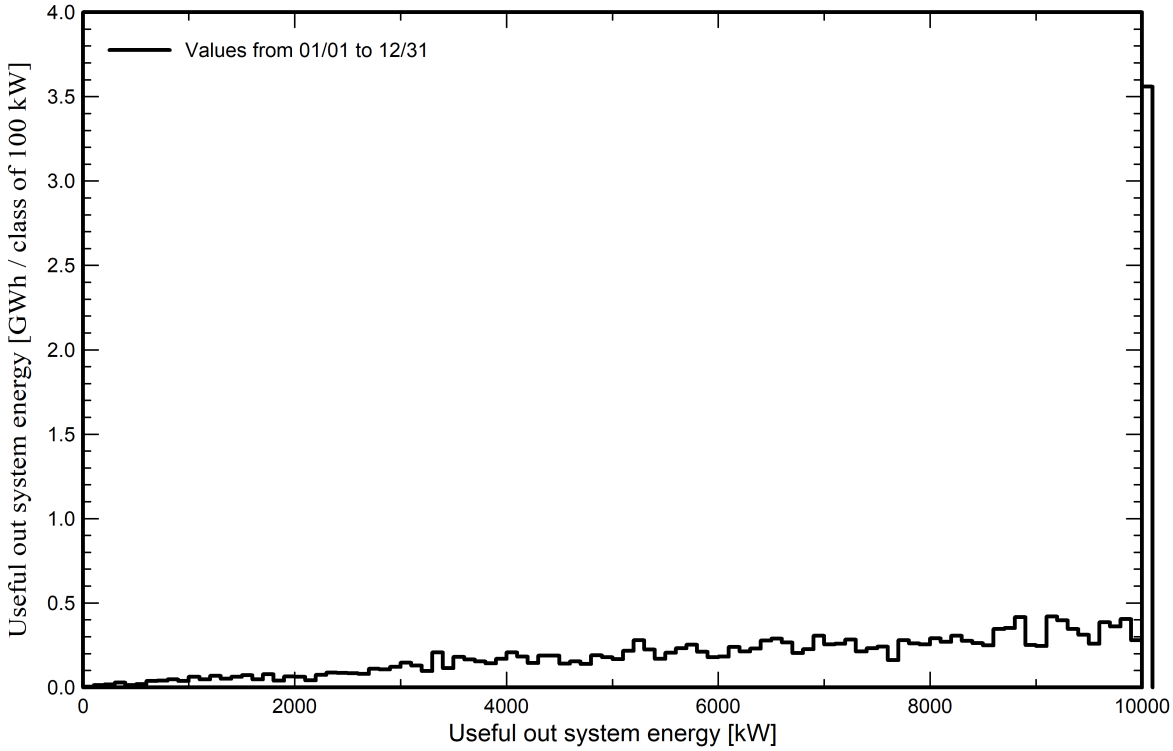


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Meteo data

Source	Meteonorm 8.1, Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	5.6 %
Specified Deviation	
Climate change	0.0 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

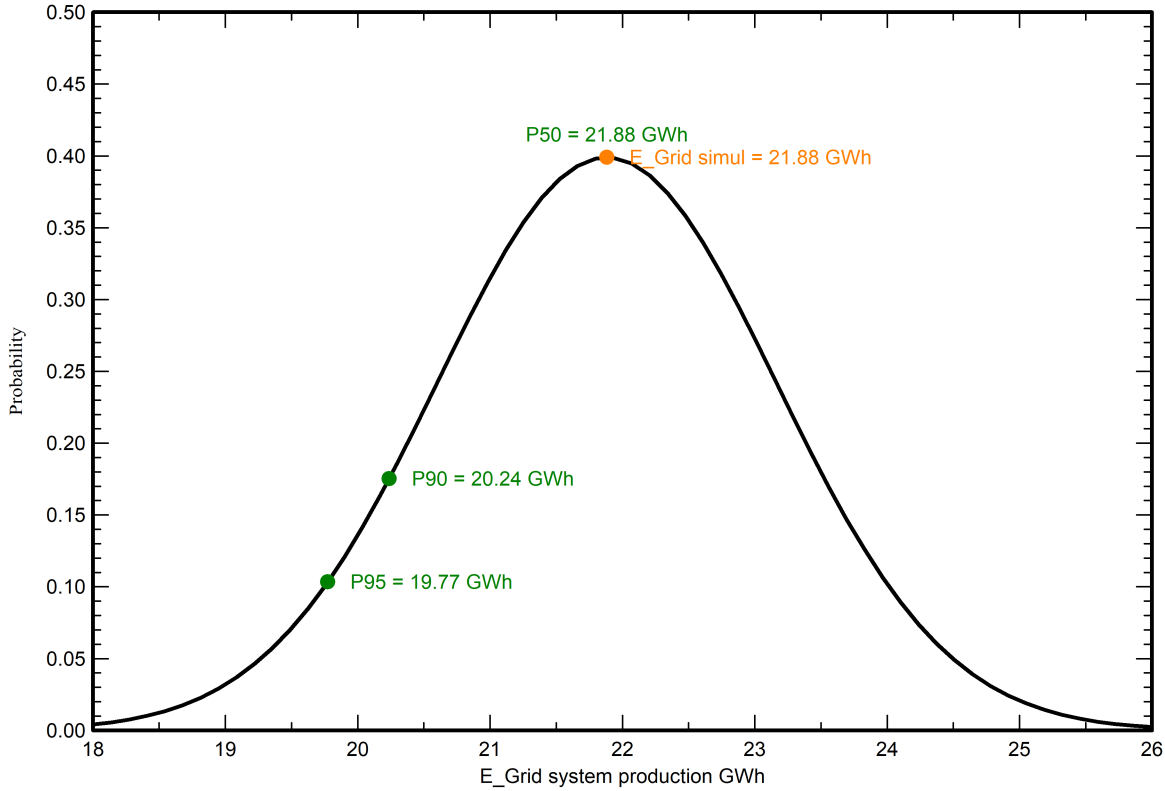
Global variability (meteo + system)

Variability (Quadratic sum)	5.9 %

Annual production probability

Variability	1.28 GWh
P50	21.88 GWh
P90	20.24 GWh
P95	19.77 GWh

Probability distribution



CO_2 Emission Balance

Total: 210931.8 tCO_2

Generated emissions

Total: 22595.36 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 269144.2 tCO_2

System production: 21881.65 MWh/yr

Grid Lifecycle Emissions: 410 gCO_2/kWh

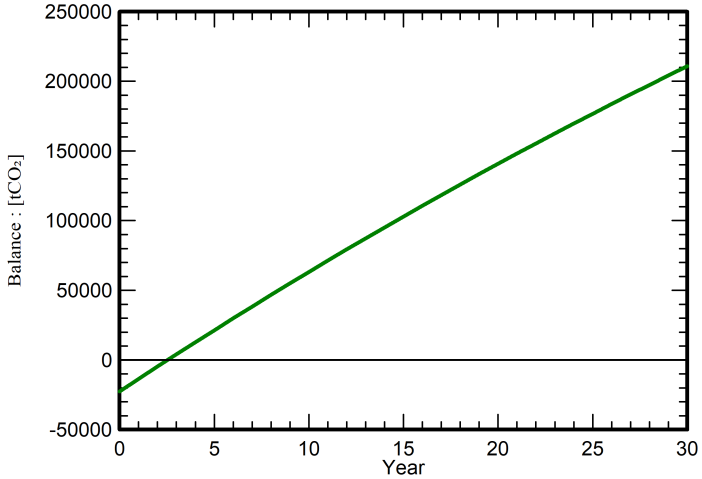
Source: IEA List

Country: Sri Lanka

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1624 kgCO2/kWp	13512 kWp	21944739
Supports	2.73 kgCO2/kg	232960 kg	637076
Inverters	271 kgCO2/	50.0	13546